Why Should They Expand?

Computational Growth, Resource Sufficiency, and Conditional Interstellar Expansion

Hyeonjun Joo

Independent Researcher, Republic of Korea

jonejoo0805@gmail.com

12 October 2026

Abstract

Technological capacity for interstellar expansion does not by itself establish a continuing incentive to expand. This paper develops the computational inward transition hypothesis (CITH) as a conditional allocation model. Recursively nested virtual worlds motivate finite-workload accounting; the allocation results apply to any internal computational use. A local service frontier separates physical feasibility from a common-budget comparison of inward investment, resident infrastructure, automated industry, and observation. Under differentiability and concavity, marginal-return inequalities characterize an all-inward optimum. Logarithmic computational value yields exact productivity thresholds and residual external-value floors. Along an unbounded-budget path, bounded relative external prices and a residual slope bounded away from zero eventually exclude the inward corner, regardless of local productivity. Persistence instead requires an appropriate price-to-residual-value path, sufficient productivity, and continued physical feasibility. A shifted-isoelastic extension shows that logarithmic value is a boundary case: slower decay of marginal computational value permits an eventual inward productivity threshold, whereas faster decay makes the inward productivity set bounded when residual slopes are positive. Concave external production preserves the corner test when initial marginal returns are unchanged, although mixed allocations change. Access-cost sensitivity and a two-period resource model expose additional routes back to outward investment. Allocation, detectability, and population-level silence remain distinct claims. The contribution is a reproducible account of conditions under which internal growth and no additional external investment can coexist, not evidence that those conditions hold for actual civilizations.

Keywords: Fermi paradox; astrobiology; computational inward transition; resource allocation; interstellar expansion; technosignatures.

1 Introduction

Expansion-based formulations of the Fermi paradox connect technological capability to persistent settlement or self-replicating activity. Hart (1975) and Tipler (1980) gave influential versions of this argument; Brin (1983) reviewed the broader assumptions behind the resulting "Great Silence." Diffusion, finite settlement lifetimes, stellar motion, and heterogeneous sites complicate the route from launch capability to complete occupation (Newman and Sagan, 1981; Carroll-Nellenback et al., 2019). Cosmological models nevertheless show why even a small number of persistently expanding lineages can matter (Olson, 2015; Hanson et al., 2021; Kipping, 2026). This paper examines a behavioural premise rather than denying these propagation arguments: does a civilization that can expand necessarily continue to find additional external investment worthwhile?

The computational inward transition hypothesis (CITH) considers continued growth directed inward. Virtual worlds may contain agents or processes that generate further worlds, producing

an approximately self-similar hierarchy of designed environments, institutions, histories, and experiences. A civilization need not become passive or lose its ambition. It may increase the quantity, variety, or quality of valued activity without repeatedly establishing resident infrastructure elsewhere. Technological progress may also make the supporting services of a chosen internal growth path sufficiently available within existing territory. The strong hypothesis combines these possibilities: *continued internal growth can coexist with no additional external investment when local support remains sufficient and the relevant allocation inequalities continue to hold.* Recursively nested worlds motivate a workload-accounting example; the allocation mechanism does not require recursion.

This is not the assertion that an ordinary finite computer implements infinitely many independent worlds at once. Nor is a computer inside a simulation an additional physical computer that can be added to the host's capacity. CITH is a production and allocation hypothesis subject to physical accounting, not a source of free substrate. Energy, state storage, entropy management, replacement material, error correction, and failure risk remain relevant (Landauer, 1961; Bennett, 1973; Lloyd, 2000). The distinction between a generative rule and its instantiated states will be explicit.

1.1 Relation to earlier work and the claimed contribution

The qualitative ideas of postbiological civilizations and inward development are not new. Dick (2003) and Ćirković and Bradbury (2006) considered computationally oriented civilizations; Smart (2012) proposed transcension toward increasingly dense internal scales. Aestivation concerns the timing of computation rather than substituting internal development for external investment (Sandberg et al., 2017). The thermodynamic critique by Bennett et al. (2019) is important here: efficiency cannot be evaluated without specifying accessible entropy reservoirs. Sustainability and homeostasis offer different mechanisms for bounded growth (Haqq-Misra and Baum, 2009; Wong and Bartlett, 2022). The present paper neither imports a universal evolutionary destiny nor infers preferences from energy efficiency alone.

A non-peer-reviewed working paper uses the phrase "Inward Transition" for a post-AGI shift toward internal computation and reduced detectability (Pitti, 2026). No priority is claimed for that phrase, virtual worlds, or inwardness. Conversely, resource acquisition and self-replicating industry can strengthen expansion incentives, including on intergalactic scales (Armstrong and Sandberg, 2013); autonomous machine industry may reduce dependence on biological settlement (Ivliev, 2026). Population-total value can make unused astronomical resources represent substantial foregone welfare (Bostrom, 2003). These are direct counterpressures, not exceptions to be excluded by definition. Social-scientific critiques also caution against treating poorly identified motives or human scaling laws as universal civilizational laws (Shkurko, 2024; Jackson and Criado-Perez, 2024).

The proposed contribution is a more specific combination: a workload frontier, a common-budget comparison of internal and external activity, exact logarithmic thresholds, a curvature trichotomy, and pathwise persistence conditions. Access costs, larger budgets, residual external value, production curvature, and future resource production supply explicit tests of the mechanism. The optimization and branching-process tools are standard. Their application identifies assumptions that a strong inward-growth claim must satisfy and circumstances in which it fails. The analytical thresholds, rather than a fraction of a synthetic ensemble, are the main result.

1.2 Four distinct claims

The strong inward-growth claim must be separated from three neighbouring statements. First, fewer new resident settlements may coexist with extensive external mining and probes; this is a weaker *settlement-composition* branch. Second, either branch could be bright or faint. Third, even common inward allocation need not explain galactic silence because autonomous

expansionist descendants can dominate territory. We distinguish the *allocation claim*, the *detectability claim*, and the *population claim*. The formal focus is all-inward allocation, not merely reduced settlement. Neither detectability nor a population-level explanation of silence follows from that allocation alone.

2 Workload accounting and local resource sufficiency

2.1 Experiential substitution does not create external facts

Let $q \in [0,1]$ represent decision-relevant experiential fidelity, with an increasing substitution map $\sigma(q) \in [0,1]$. If $E(n)$ is the experiential benefit of additional residence before substitution, and $X(n)$ is residual value, write

$$B_{\text{phys}}(n,q) = [1-\sigma(q)]E(n) + X(n). \tag{1}$$

At perfect substitution only the first term disappears. Unknown independent histories, raw material, strategic options, or a preference for physical authenticity may remain valuable. For a discrete project with substitutable value e_j, residual value x_j, and incremental cost c_j relative to an internal alternative,

$$\Delta_j(q) = [1-\sigma(q)]e_j + x_j - c_j. \tag{2}$$

A positive reality premium is compatible with rejecting the project when it does not cover its extra cost. This is not an assertion of metaphysical identity between a simulated and an external object.

The reduced-form q is domain-specific. It does not measure consciousness, establish the moral equivalence of digital minds, or imply that simulated observations reveal unknown external facts. An embodied population using virtual environments is sufficient for the weaker experiential interpretation. The recursive digital-population interpretation additionally assumes that the relevant minds can be realized computationally.

2.2 Generative depth and implemented workload

Consider a regular illustrative hierarchy with integer branching factor $b \geq 2$ and maximum integer depth $d \geq 0$. Its number of represented worlds is

$$W_d = \sum_{j=0}^{d} b^j = \frac{b^{d+1}-1}{b-1}. \tag{3}$$

A recursive rule can lack a prescribed terminal depth even though every actual execution has finite depth, finite state, and finite elapsed computation. Approximate self-similarity is an architectural description; no physical fractal dimension or infinite computational speed-up is inferred from it. A hierarchy need not be regular in applications, and counts alone do not measure novelty or welfare.

For a fully instantiated hierarchy, let $u_k > 0$ be the normalized per-world requirement for physical service k, $\eta_k > 0$ a stated efficiency factor, and K_k the locally available capacity in matching units. Examples of k include update work during an interval, persistent memory, heat rejection, and material replacement. The simplifying homogeneous accounting is

$$D_k(d) = \frac{u_k W_d}{\eta_k}, \qquad D_k(d) \leq K_k \quad \text{for every } k. \tag{4}$$

These are separate constraints, not quantities to be added without a common conversion. For cumulative energy or material consumption, D_k and K_k must refer to integrated demand and the available stock over the same horizon. Instantaneous power cannot be compared with lifetime energy.

Lemma 1 (Finite-depth support frontier)**.** *Under equation* (4)*, define* $Q = \min_k K_k\eta_k/u_k$*. If* $Q \geq 1$*, the largest feasible integer depth is*

$$d_{\max} = \lfloor \log_b[1 + (b-1)Q] - 1 \rfloor . \tag{5}$$

If $Q < 1$*, even the root world at the specified workload is infeasible. Finite positive* $K_k\eta_k/u_k$ *for at least one service imposes a finite depth ceiling.*

Proof. Every service constraint is equivalent to $W_d \leq K_k\eta_k/u_k$. Their intersection is $W_d \leq Q$. Substitution of equation (3), followed by taking a logarithm and restricting d to integers, gives equation (5). □

The lemma does not forbid continued creation of worlds. It specifies the trade-off. Deeper simultaneous hierarchies require more effective support, less per-world workload, or a different execution schedule. The u_k parameters include whatever fidelity, update rate, individuality, addressing, and coordination the stipulated worlds require. Compression is not assumed to remove independent state information without loss.

A scheduled hierarchy requires a different account. Processing and heat may depend on the active set $A(t) \leq W_{d(t)}$, but preserved inactive states still consume memory. Reusing a program description does not remove accumulated histories. The numerical example below therefore charges processing and heat to the active worlds and persistent memory to *all* represented world states. Scheduling changes simultaneity and completion time; it is not a demonstration that arbitrary independent worlds can all be lived at full speed with constant resources.

What recursion adds. For the fully instantiated hierarchy at fixed per-world workload and efficiency, equation (3) gives $W_{d+1} = bW_d + 1$, and hence

$$\frac{D_k(d+1)}{D_k(d)} = b + \frac{1}{W_d} \longrightarrow b. \tag{6}$$

If useful computation is proportional to fully instantiated workload, $C_d = cW_d$ with fixed $c > 0$, logarithmic value adds only

$$F(C_{d+1}) - F(C_d) = \ln\left[\frac{1 + c(bW_d + 1)}{1 + cW_d}\right] \longrightarrow \ln b. \tag{7}$$

Both limits follow by dividing numerator and denominator by W_d and using $W_d \to \infty$. Thus deeper nesting is not a computational free lunch. Recursion supplies a concrete generative workload, not a distinct preference mechanism: replacing it by any non-recursive internal workload with the same service demands and computational value leaves the allocation results in sections 3–5 unchanged.

2.3 Support over a development path

For a specified horizon $[0, T]$, local sufficiency means

$$\chi_T = \sup_{0 \leq t \leq T} \max_k \frac{D_k(t)}{K_k(t)} \leq 1. \tag{8}$$

Reliability also requires an explicit service criterion; extra local copies need not remove common-mode failures. Technological progress can reduce u_k, increase usable η_k or K_k, improve repair, or alter the execution schedule. These are possible pathways, not measured capabilities. In particular, reversible computation does not authorize neglecting state preparation, error management, or the entropy reservoirs actually available (Bennett, 1973; Bennett et al., 2019).

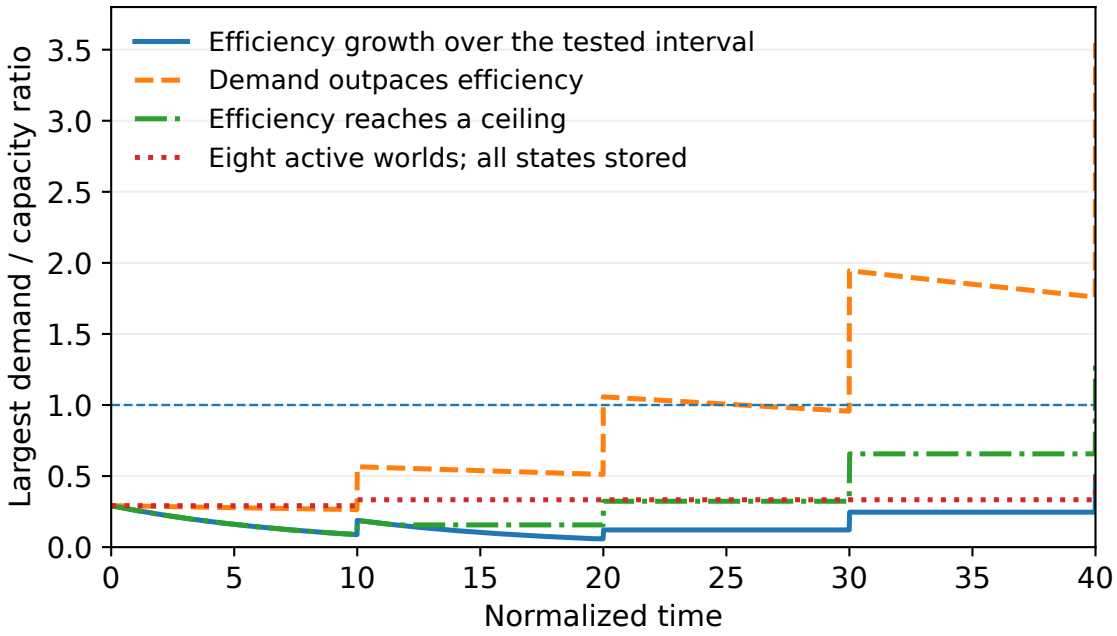


Figure 1: Finite-horizon local support constructions. The largest service-demand ratio must remain at or below one. Processing and heat are charged to active worlds; persistent memory is charged to all represented states, including inactive ones. Growth in depth is stepwise. No curve is extrapolated to unlimited time, and no physical implementation is validated.

A simple rate comparison clarifies the temporal issue. If a service ratio evolves as $\chi_k(t) = \chi_{k0} \exp[(g_{D,k} - g_{K,k})t]$, it crosses one at

$$T_k = \frac{\ln(1/\chi_{k0})}{g_{D,k} - g_{K,k}} \tag{9}$$

when $0 < \chi_{k0} < 1$ and demand growth exceeds capacity growth. A finite-horizon success does not imply sufficiency at every future time. Conversely, external acquisition is not required by this accounting over an interval on which all demands are met locally.

Four controlled examples use $b = 2$, $d(t) = 2 + \lfloor t/10 \rfloor$ on $0 \le t \le 40$, and local capacities $(32, 256, 24)$ for update work, stored-state memory, and heat rejection. All $u_k = 1$. Processing and thermal efficiency either grow as $e^{0.12t}$, grow as $e^{0.01t}$, or grow as $e^{0.12t}$ up to a ceiling of four; memory efficiency remains one. A fourth example activates at most eight worlds at a time without efficiency improvement. Every example stores all W_d states. These are normalized accounting constructions, not device specifications.

The matched-efficiency example remains feasible while the full hierarchy grows from seven to 127 worlds. Demand outpacing efficiency first violates a constraint at $t = 20$, and the efficiency-ceiling example first violates one at $t = 40$. Scheduling remains feasible on the tested interval, but only eight worlds are active at once. The maximum ratios are respectively 0.4961, 3.5471, 1.3229, and 0.4961. All changes in depth occur at known times, so interval maxima can be checked at the jumps rather than inferred from a sampled time grid alone. Figure 1 illustrates the distinction between technology-supported expansion of internal activity and merely postponing a resource limit. Extending the matched-efficiency path with the same fixed memory efficiency and capacity gives $W = 255$ and a memory ratio $255/256 = 0.99609375$ at $t = 50$, but $W = 511$ and $511/256 = 1.99609375$ at $t = 60$. The latter is its first memory violation. The example therefore supports the stated finite horizon, not indefinite local sufficiency.

3 Allocation between inward growth and external projects

Feasibility and optimality are different questions. Even a locally supportable civilization may prefer to acquire further resources. To test that preference without removing the internal alternative, let $z, x, y, w \geq 0$ denote intensive local investment, additional resident infrastructure, industry-only capacity, and observation-only capacity. With local investment as numeraire, the common budget is

$$z + p_n x + p_h y + p_m w \leq B, \qquad p_j > 0, \quad B > 0. \tag{10}$$

All quantities are normalized within one planning horizon. The cumulative indices $n = 1 + x$, $h = 1 + x + y$, and $m = 1 + x + y + w$ distinguish residence, industry, and observation. They are continuous service-capacity indices in this model, not literal fractional star systems. The nesting is an accounting assumption, not a necessary geometry of every civilization. Their use does not assume that star systems themselves are divisible.

Consider

$$C = C_0 + a \ln(1 + z) + r_n x + r_h y, \tag{11}$$

$$U = F(C) + (1 - q)E(x) + \epsilon_R R(x + y) + \epsilon_I I(x + y + w), \tag{12}$$

where $a > 0$, $C_0 \geq 0$, $r_n, r_h \geq 0$, and the residual weights are non-negative. The specialization $\sigma(q) = q$ is used here. C is useful computational service over the horizon, not a count of simulated hardware added to real hardware. E is substitutable settlement-specific experience; R and I represent residual resilience and independent-information value. The common budget prevents counting external investment as beneficial without comparing what the same resources could produce internally.

A small ϵ_R or ϵ_I means low residual marginal value in the objective. Hard energy, reliability, or material requirements are separate feasibility constraints, as in equation (8); the sensitivity scenarios therefore use the label *large residual values*.

3.1 An all-inward optimum without bounded total value

Assume F is differentiable, strictly increasing, and concave, and E, R, I are differentiable, non-decreasing, and concave on the feasible domain. Composition with the concave C makes U concave. Positive a and increasing F exhaust the budget. At the all-inward corner $(z, x, y, w) = (B, 0, 0, 0)$ define

$$C_L = C_0 + a \ln(1 + B), \qquad \lambda_L = F'(C_L)\frac{a}{1 + B}, \tag{13}$$

$$D_n = F'(C_L)r_n + (1 - q)E'(0) + \epsilon_R R'(0) + \epsilon_I I'(0),$$
$$D_h = F'(C_L)r_h + \epsilon_R R'(0) + \epsilon_I I'(0), \qquad D_m = \epsilon_I I'(0). \tag{14}$$

Proposition 2 (All-inward allocation)**.** *For equations* (10) *and* (12) *under the stated differentiability and concavity assumptions, the all-inward corner is globally optimal if and only if*

$$\frac{D_j}{p_j} \leq \lambda_L \qquad (j = n, h, m). \tag{15}$$

If all three inequalities are strict, every optimizer has zero external investment. Equality can admit ties; uniqueness is not asserted from a weak inequality.

Proof. For any feasible displacement from the corner, the supporting-hyperplane inequality gives

$$\begin{aligned} U(z, x, y, w) - U(B, 0, 0, 0) &\leq \lambda_L(z - B) + D_n x + D_h y + D_m w \\ &\leq (D_n - p_n\lambda_L)x + (D_h - p_h\lambda_L)y + (D_m - p_m\lambda_L)w. \end{aligned} \tag{16}$$

The right side is non-positive under equation (15), and strictly negative for any positive external component if all inequalities are strict. Conversely, transferring a sufficiently small budget amount from z to a violating external direction has positive directional derivative and improves the objective. □

This is the standard supporting-hyperplane or first-order optimality criterion for a concave maximization problem (Boyd and Vandenberghe, 2004, chapters 3–5), specialized to the civilizational allocation model. It does not require bounded total ambition: $F(C) = \ln(1 + C)$ grows without bound as $C \to \infty$. It does require specified alternatives, prices, residual preferences, and an admissible local path. Its contribution is not a new general optimization theorem, but the identification of the competing terms relevant to the hypothesis.

Corollary 3 (Adding hard support constraints)**.** *Suppose an additional physical feasibility restriction retains the all-inward corner. If equation* (15) *holds, the corner remains optimal in the restricted problem. The inequalities need not remain necessary: at a binding physical constraint, a formerly improving external direction may no longer be feasible. Necessity is recovered when each small budget-transfer direction used in the converse proof remains admissible.*

Proof. A maximum over the larger feasible set is a maximum over any subset that contains it. The converse uses only the feasibility of the indicated local directions. □

The corollary prevents a numerical preference calculation from being mistaken for engineering validation. An all-inward optimum with $\chi_T > 1$ is not an implementable explanation of the chosen workload. An outward optimum despite $\chi_T \leq 1$ describes attractive expansion, not expansion forced by a local shortfall. The two tests must be reported separately.

3.2 Exact thresholds and computational-value curvature

Specialize to $F(C) = \ln(1 + C)$. Write the residual slopes as

$$\begin{aligned} S_n &= (1-q)E'(0) + \epsilon_R R'(0) + \epsilon_I I'(0), \\ S_h &= \epsilon_R R'(0) + \epsilon_I I'(0), \qquad\qquad S_m = \epsilon_I I'(0), \end{aligned} \tag{17}$$

and set $r_m = 0$, $L = \ln(1 + B)$ and $d_0 = 1 + C_0$. Define

$$H_j = (1 + B)LS_j, \qquad A_j = (1 + B)(r_j + d_0 S_j). \tag{18}$$

Direct substitution into equation (15) gives the exact equivalent test

$$\boxed{p_j \geq H_j + \frac{A_j}{a} \quad \text{for all } j.} \tag{19}$$

No numerical optimizer is needed to determine whether this corner is optimal.

Proposition 4 (Productivity threshold and residual floor)**.** *Hold B, C_0, p_j, r_j, S_j fixed. Assume at least one $A_j > 0$; if none is positive every $a > 0$ is inward. For every direction with $A_j > 0$, a finite productivity level satisfying equation* (19) *exists if and only if $p_j > H_j$. When all relevant denominators are positive,*

$$a \geq a_{\text{crit}} := \max_{j:A_j>0} \frac{A_j}{p_j - H_j} \tag{20}$$

is necessary and sufficient. A direction with $A_j = 0$ imposes no restriction because then $r_j = S_j = H_j = 0$. If any $A_j > 0$ has $p_j \leq H_j$, no finite increase of a alone produces an all-inward corner.

Proof. All factors used to obtain equation (19) are positive. Rearranging $A_j/a \leq p_j - H_j$ yields the result; the zero case follows from non-negativity of its constituent terms. □

The quantity H_j is a *residual-value floor relative to local marginal return*, not an irreducible energy cost or a physical impossibility theorem. It shows precisely why "better computers" alone can fail: external information or other residual value may remain competitive even when remote computational value becomes relatively small. Improving substitution or local support can alter these residual preferences, but their disappearance must not be silently assumed.

A dimensionless robustness diagnostic is

$$\rho = \max_j \frac{H_j + A_j/a}{p_j}. \tag{21}$$

The corner is inward when $\rho \le 1$. If every external price is multiplied by $\kappa > 0$ while other terms are held fixed, the corner survives exactly when $\kappa \ge \rho$. Thus reporting ρ identifies how much a conclusion depends on proximity to a threshold. This is conditional price sensitivity, not a distribution over unknown technologies.

Curvature generalization. Use the normalized shifted-isoelastic family

$$F_\eta(C) = \begin{cases} [(1+C)^{1-\eta} - 1]/(1-\eta), & \eta \neq 1, \\ \ln(1+C), & \eta = 1, \end{cases} \qquad \eta \ge 0. \tag{22}$$

Thus $F'_\eta(C) = (1+C)^{-\eta}$, $F_\eta(0) = 0$ and $F'_\eta(0) = 1$. The restriction $\eta \ge 0$ retains concavity; negative η is outside the global corner theorem. The curvature parameter η is distinct from the service-efficiency factors η_k.

Proposition 5 (Curvature trichotomy for inward allocation)**.** *Fix $B > 0$, $d_0 \ge 1$, $p_j > 0$ and finite $r_j, S_j \ge 0$, and set $L = \ln(1+B)$. Under the common-budget model and its concavity assumptions, a direction satisfies the all-inward test exactly when*

$$r_j + S_j(d_0 + aL)^\eta \le \frac{p_j a}{1+B}. \tag{23}$$

For $0 \le \eta < 1$, every nontrivial direction has a unique finite threshold, and the all-inward corner is optimal for every a above the largest of those thresholds. For $\eta = 1$, the exact condition is $p_j \ge H_j + A_j/a$; a finite threshold exists if and only if $p_j > H_j$ in every nontrivial direction. For $\eta > 1$, the inward set in $a > 0$ for each direction with $S_j > 0$ is either empty, a singleton, or a bounded closed interval. Consequently, if any direction has $S_j > 0$, sufficiently high productivity precludes an all-inward corner. Directions with $S_j = 0$ retain the threshold $a \ge (1+B)r_j/p_j$ for every η; a direction with $r_j = S_j = 0$ is unrestricted.

Proof. At the corner, $1 + C_L = d_0 + aL$. The local marginal return is $a/[(1+B)(d_0+aL)^\eta]$, and direction j has marginal return $r_j/(d_0+aL)^\eta + S_j$. The first-order condition from Proposition 2, multiplied by the positive quantity $(d_0+aL)^\eta$, gives (23). Since F_η is increasing and concave for $\eta \ge 0$, that condition remains necessary and sufficient for global optimality when it holds in all directions.

For a nontrivial direction define its required relative price

$$P_{j,\eta}(a) = \frac{1+B}{a}[r_j + S_j(d_0 + aL)^\eta].$$

For $0 \le \eta < 1$, differentiation gives

$$P'_{j,\eta}(a) = \frac{1+B}{a^2}\left\{-r_j + S_j(d_0+aL)^{\eta-1}[(\eta-1)aL - d_0]\right\} < 0.$$

The strict inequality follows from nontriviality, $d_0 > 0$ and $\eta < 1$. Moreover, $P_{j,\eta}(a) \to \infty$ as $a \downarrow 0$, whereas $P_{j,\eta}(a) \to 0$ as $a \to \infty$: its two terms are of orders a^{-1} and $a^{\eta-1}$, respectively.

Continuity and strict monotonicity imply exactly one solution of $P_{j,\eta}(a) = p_j$, and the direction passes precisely above it. Taking the maximum over finitely many directions proves the first case. At $\eta = 1$, expansion of the bracket gives $P_{j,1}(a) = H_j + A_j/a$, proving the second case, including the strictly positive denominator requirement.

For $\eta > 1$ and $S_j > 0$, put

$$D_j(a) = r_j + S_j(d_0 + aL)^\eta - \frac{p_j a}{1+B}.$$

Then $D_j(0) = r_j + S_j d_0^\eta > 0$ and

$$D_j''(a) = S_j \eta(\eta - 1)L^2(d_0 + aL)^{\eta-2} > 0.$$

Also $D_j(a) \to \infty$ as $a \to \infty$, because its positive a^η term dominates its linear term. Its non-positive sublevel set is convex by convexity, closed by continuity, bounded by this divergence, and bounded away from zero by $D_j(0) > 0$. It is therefore empty, a singleton, or a bounded closed interval contained in $(0, \infty)$. The intersection over directions is bounded if any such direction exists. Finally, with $S_j = 0$, (23) reduces directly to the stated threshold, including the unrestricted zero-return case. □

The finite, positive high-productivity price limit is specific to $\eta = 1$: it is zero for $\eta < 1$, equals H_j for $\eta = 1$, and diverges for $\eta > 1$ when $S_j > 0$. Thus it would be incorrect to say that only logarithmic value allows residual benefits to defeat increases in productivity. The distinction is between a finite asymptotic floor and an unbounded high-productivity requirement. The $\eta > 1$ branch has bounded total computational value; the $0 \le \eta \le 1$ branches do not.

For the normalized example $B = 10$, $r = 0.1$, $S = 0.001$, $d_0 = 2$, $p = 1$ and $\eta = 2$, (23) is a quadratic inequality. The quadratic roots are $a_- \simeq 1.421897$ and $a_+ \simeq 12.720521$, with inward allocation exactly on $[a_-, a_+]$. Separate scalar maximizations of a concave objective with residual utility $S(1 - e^{-x})$ verify inward solutions at $a = 2$ and $a = 10$, and outward solutions at $a = 1$ and $a = 20$.

3.3 A joint finite-interval witness

The support and allocation examples need not be merely disconnected illustrations. Couple the matched-efficiency schedule in section 2 to $B = 10$, $C_0 = 0$, $a(t) = W_{d(t)}/\ln(11)$, $r_n = r_h = 0.1$, and all three prices equal to one. Set $q = 0.99$, $\epsilon_R = \epsilon_I = 0.001$, with unit initial slopes for E, R, I. At $z = B$, useful computation is exactly $C_L = W_{d(t)}$, the workload charged in the support example. The three outward-to-local marginal-return ratios are

$$11\ln(11)\left[S_j + \frac{r_j + S_j}{W_{d(t)}}\right], \qquad (S_n, S_h, S_m) = (0.012, 0.002, 0.001). \tag{24}$$

They decrease with world count and are already below one at seven worlds. Thus this same finite family satisfies both the three-service feasibility test and strict all-inward optimality with positive residual external values. As the hierarchy grows from seven to 127 worlds, attained logarithmic value rises from $\ln 8$ to $\ln 128$. The budget is fixed at $B = 10$ and productivity is deliberately tuned to $a(t) = W_{d(t)}/\ln 11$. This is a consistency witness, not evidence for the plausibility of that productivity path, an engineering implementation, or an infinite-horizon optimum. Persistent memory is still charged to every world; future resource effects are considered separately in section 5.

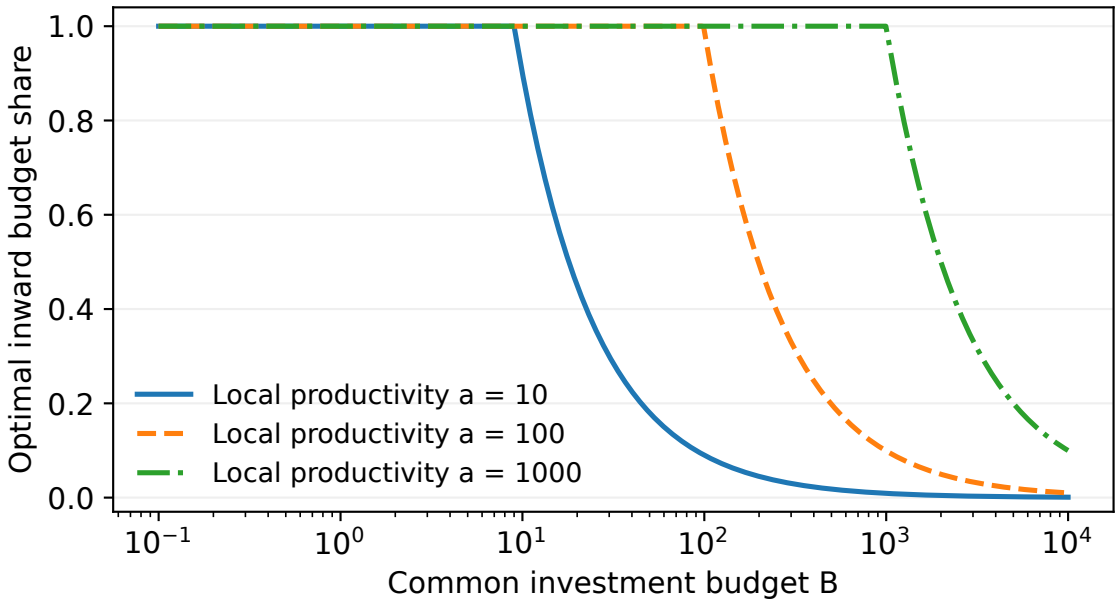


Figure 2: Exact inward budget share with perfect experiential substitution, zero residual value, and $R_* = 1$. The transition occurs at $B = a - 1$. Greater local productivity moves the boundary, but budget growth can eventually favor additional external capacity. Any strictly increasing computational value function gives the same allocation in this pure-computation case.

3.4 Budget growth and shared technological progress

With perfect experiential substitution and no residual external value, $q = 1$ and $\epsilon_R = \epsilon_I = 0$. For *any* strictly increasing F, maximizing value is equivalent to maximizing C. Let

$$R_* = \max\{r_n/p_n, r_h/p_h\}. \tag{25}$$

When $R_* > 0$, the solution allocates

$$z^* = \min\{B, \max(a/R_* - 1, 0)\}, \qquad B_{\text{crit}} = a/R_* - 1. \tag{26}$$

Any remaining budget goes to a best-return computational external channel. The all-inward condition is $a/(1 + B) \geq R_*$. If $R_* = 0$, all investment is local. This exposes a limitation of a fixed-budget productivity path: with fixed a and $R_* > 0$, a sufficiently large budget restores external investment even without authenticity, information, or resilience premiums. The civilization does not stop preferring more value when it can generate virtual worlds.

If local and remote computational coefficients all receive the same multiplicative improvement, the pure-computation condition is unchanged. Progress that disproportionately raises intensive return can support inward allocation; progress that improves everything equally need not. Figure 2 shows this budget effect for $R_* = 1$. Persistent inward allocation along a development path requires the relevant inequalities to hold along that path, not merely at one chosen value of B.

Persistence when budgets grow. For logarithmic computational value, let $d_0 = 1 + C_0 \geq 1$ and $r_j \geq 0$ be fixed. At time t, define

$$L(t) = \ln[1 + B(t)], \qquad H_j(t) = [1 + B(t)]L(t)S_j(t), \qquad A_j(t) = [1 + B(t)][r_j + d_0 S_j(t)].$$

All quantities below are finite at each finite time. Prices are relative to local investment, the numeraire.

Proposition 6 (Pathwise persistence with logarithmic value)**.** *Suppose $B(t) > 0$, $a(t) > 0$, $p_j(t) > 0$ and $S_j(t) \geq 0$ on a path $t \geq t_0$, and retain the differentiability and concavity assumptions of Proposition 2. The all-inward corner is optimal at every time on this path if and only if, for every t and every direction with $A_j(t) > 0$,*

$$p_j(t) > H_j(t), \qquad a(t) \geq \max_{j:A_j(t)>0} \frac{A_j(t)}{p_j(t) - H_j(t)}. \tag{27}$$

The maximum over an empty set is zero. In particular, wherever $S_j(t) > 0$,

$$\frac{p_j(t)}{S_j(t)} > [1 + B(t)] \ln[1 + B(t)]. \tag{28}$$

If $B(t) \to \infty$, a bounded price and a residual slope bounded away from zero in even one direction preclude eventual all-inward allocation, regardless of $a(t)$. With a fixed positive residual slope, persistence requires its relative external price to grow at least on the order of $B(t) \ln B(t)$. With bounded prices, it instead requires $S_j(t) = O(1/[B(t) \ln B(t)])$ for each direction. These order conditions are necessary, not sufficient: the productivity inequality in (27) *must also hold. If $S_j(t) = 0$ but $r_j > 0$, that inequality reduces to $a(t) \geq [1 + B(t)]r_j/p_j(t)$.*

Proof. At each time, the exact logarithmic corner test is $p_j(t) \geq H_j(t) + A_j(t)/a(t)$. If $A_j(t) > 0$ and $a(t)$ is finite and positive, this requires $p_j(t) > H_j(t)$; rearrangement gives the stated lower bound on $a(t)$. Conversely these two inequalities imply the exact test. If $A_j(t) = 0$, non-negativity and $d_0 > 0$ imply $r_j = S_j(t) = H_j(t) = 0$, so that direction imposes no restriction. This proves the pointwise equivalence and therefore the pathwise one. For $S_j(t) > 0$, we have $A_j(t) > 0$, and division of $p_j(t) > H_j(t)$ by $S_j(t)$ gives (28). Set $G(B) = (1 + B) \ln(1 + B)$. Then $G(B) \to \infty$ and $G(B)/(B \ln B) \to 1$ as $B \to \infty$. If $p_j(t) \leq M$ and $S_j(t) \geq s > 0$ eventually, $H_j(t) \geq sG(B(t)) > M$ eventually, contradicting (27). The same ratio inequality gives $p_j(t) > S_j G(B(t))$ when the slope is fixed, and $S_j(t) < M/G(B(t))$ when prices are bounded, including any zero-slope times. Finally, setting $S_j(t) = 0$ in the exact corner test gives the last assertion. □

Proposition 6 characterizes a sequence of static optima. In general, its conditions are neither necessary nor sufficient for an all-inward infinite-horizon optimal policy. The two-period examples in section 5 demonstrate the failure of both implications already when only one continuation period is added. With additional hard support constraints, necessity requires the small budget-transfer directions used in Proposition 2 to remain feasible; sufficiency is inherited whenever the corner remains feasible.

3.5 Production asymmetry and concave external alternatives

The local term $a \ln(1 + z)$ describes intensive improvement of an existing substrate with diminishing marginal returns. Linear external terms describe replication into additional territory over the investment range being modelled. This intensive–extensive distinction is a modelling choice, not a universal technological law. In particular, the unbounded logarithm is not a physical finite-capacity constraint; such constraints are imposed separately by the service frontier. External projects may also have diminishing returns.

Lemma 7 (Production curvature and the corner test)**.** *Replace useful computation by*

$$C = C_0 + a\ell(z) + G_n(x) + G_h(y),$$

where ℓ is differentiable, increasing and concave with $\ell(0) = 0$ and $\ell'(B) > 0$, and G_n, G_h are differentiable, non-decreasing and concave with $G_j(0) = 0$, $G_j'(0) = r_j$. Retain the original

residual utilities, set $r_m = 0$, and let $F(C) = \ln(1+C)$. The all-inward corner is globally optimal if and only if

$$p_j \geq \frac{r_j + d_0 S_j}{a\ell'(B)} + S_j \frac{\ell(B)}{\ell'(B)} \qquad (j = n, h, m). \tag{29}$$

For $\ell(z) = \ln(1+z)$ this is exactly the original logarithmic threshold, regardless of the curvature of G_n, G_h away from zero.

Proof. The modified C is concave, and composition with the increasing concave logarithm preserves concavity of the full objective. At the corner, $1 + C_L = d_0 + a\ell(B)$, the local marginal return is $a\ell'(B)/(d_0 + a\ell(B))$, and the external marginal return in direction j is $r_j/(d_0 + a\ell(B)) + S_j$. The supporting-hyperplane argument of proposition 2 applies unchanged. Multiplying its marginal-return inequality by the positive factor $(d_0 + a\ell(B))/(a\ell'(B))$ gives

$$p_j \geq \frac{r_j + S_j[d_0 + a\ell(B)]}{a\ell'(B)},$$

which is equation (29). For the local logarithm, $\ell(B)/\ell'(B) = (1+B)\ln(1+B)$ and $1/\ell'(B) = 1 + B$, giving the stated specialization. □

Corollary 7.1 (A production-independent lower bound on the residual floor)**.** *Retain the value, external-production and residual-utility assumptions of Lemma 7. Let ℓ be increasing and concave on $[0, \infty)$, with $\ell(0) = 0$, and differentiable at the positive corner budget B, where $\ell'(B) > 0$. Differentiability at zero is not required. Then the residual floor satisfies*

$$H_j^\ell(B) := S_j \frac{\ell(B)}{\ell'(B)} \geq S_j B.$$

For $S_j > 0$ and finite $a > 0$, an all-inward optimum requires $p_j > H_j^\ell(B) \geq S_j B$. Consequently, along a path $B(t) \to \infty$, a bounded relative external price and a residual slope bounded away from zero in any direction exclude eventual all-inward static allocation, regardless of productivity. A fixed positive slope requires its relative price to grow at least linearly in $B(t)$; bounded prices require $S_j(t) = O(1/B(t))$. These are necessary, not sufficient, growth conditions. The faster $B \ln B$ floor for local logarithmic production is a particular case, not a general rate.

Proof. The supporting-line inequality for the concave function ℓ, applied at B and evaluated at zero, gives

$$0 = \ell(0) \leq \ell(B) + \ell'(B)(0 - B).$$

Since $\ell'(B) > 0$, division and multiplication by $S_j \geq 0$ give $H_j^\ell(B) \geq S_j B$. The proof of Lemma 7 needs local differentiability only at $z = B$: the objective is concave on the whole feasible set, and its derivatives in the feasible directions from the inward corner exist there. Thus equation (29) also holds under the present endpoint regularity. When $S_j > 0$, its term $(r_j + d_0 S_j)/(a\ell'(B))$ is strictly positive, because $d_0 > 0$, $r_j \geq 0$ and a is finite and positive. This proves $p_j > H_j^\ell(B) \geq S_j B$.

If eventually $p_j(t) \leq M$ and $S_j(t) \geq s > 0$, then $S_j(t)B(t) > M$ eventually, contradicting the necessary inequality. For fixed $S_j > 0$, the same inequality gives $p_j(t) > S_j B(t)$. For bounded prices it gives $S_j(t) < M/B(t)$ wherever $S_j(t) > 0$, with the bound immediate at zero-slope times. Neither consequence controls the remaining productivity term in equation (29), so it does not establish sufficiency. For $\ell(B) = \ln(1+B)$, direct substitution gives $\ell(B)/\ell'(B) = (1+B)\ln(1+B)$, recovering the logarithmic rate. □

For example, $G_j(u) = r_j \ln(1 + \kappa_j u)/\kappa_j$ with $\kappa_j > 0$ is concave and has the same initial slope as $r_j u$. It preserves both the logarithmic corner test and the growth-path obstruction in proposition 6. The isoelastic corner test likewise uses only $G_j'(0)$, so proposition 5 is unchanged.

Linearity is instead needed for the specific mixed-allocation formula in equation (26), the corresponding curves in figure 2, and the closed-form continuation allocation in section 5.

A one-channel pure-computation example makes the distinction explicit. Set $a = 10$, $r = p = 1$. With $G(x) = x$, the optimum has $x^* = \max\{0, B - 9\}$. With $G(x) = \ln(1 + x)$, differentiating $10 \ln(1 + B - x) + \ln(1 + x)$ gives $-10/(1 + B - x) + 1/(1 + x)$; strict concavity therefore gives $x^* = \max\{0, (B - 9)/11\}$. Both cases leave the inward corner at the same budget boundary, but their positive external allocations differ. The executable checks compare both formulas with scalar optimization. The particular $B \ln B$ rate in proposition 6 comes from the local logarithmic production function, not from external linearity.

4 Access-cost sensitivity

Reducing external prices by a factor of 100 eliminates every all-inward corner in the specified 1,024-input reference design: the count falls from 219 to zero. This controlled experiment holds computational returns, residual utilities, and local productivity fixed. The result is a sensitivity of this design, not a universal critical discount. It places the central mechanism in perspective: inward allocation is compatible with rapid access, but is not generally robust to sufficiently cheap external opportunities.

4.1 Analytical phase boundary

For logarithmic value, define

$$\xi_j = \frac{H_j}{p_j}, \qquad \zeta_j = \frac{A_j}{p_j a}. \tag{30}$$

The exact phase boundary is $\xi_j + \zeta_j = 1$. All-inward allocation requires $\xi_j + \zeta_j \leq 1$ for *every* direction. If $\xi_j \geq 1$, no finite productivity restores the corner, because productivity lowers only ζ_j. Here $\xi_j \geq 1$ entails $S_j > 0$ and hence $A_j > 0$, so ζ_j remains strictly positive at every finite a. Increasing a to λa, with $\lambda > 1$, moves (ξ_j, ζ_j) vertically to $(\xi_j, \zeta_j/\lambda)$. Reducing the external price to p_j/κ, with $\kappa > 1$ and other parameters fixed, moves it radially to $(\kappa\xi_j, \kappa\zeta_j)$. The degenerate case $A_j = 0$ has $H_j = 0$ and imposes no restriction. Figure 3 shows this analytical geometry rather than an ensemble frequency.

4.2 Paired sensitivity design

To distinguish deployment costs from access delays, use the normalized family

$$p_j = p_{j0}\left(f + \frac{1}{s_t} + \frac{0.2}{s_c}\right), \qquad r_j = r_{j0} m_r \exp(-0.1/s_t), \qquad a = a_0 m_a. \tag{31}$$

Here $f > 0$ is the residual price factor; $s_t, s_c > 0$ index effective material and information access speeds relative to c. At instantaneous access, $p_j = p_{j0} f$ and $r_j = r_{j0} m_r$. The coefficients 0.2 and 0.1 define scenarios, not calibrated physical laws. Speed sweeps also remove the stipulated attenuation of remote returns, so they are *not* pure price experiments. The fixed-access sweeps in f isolate the price effect.

Faster-than-light access is only a counterfactual limiting case of this map, not a physical mechanism established here. Instantaneous arrival still leaves the positive deployment price $p_{j0} f$. Exactly zero external prices are excluded: they can destroy compactness of the feasible set and, with positive external computational returns and unbounded increasing value, need not admit a finite optimizer.

The reference design contains 1,024 scrambled Sobol points in 14 dimensions, paired across every scenario. The exact library, generator call and frozen inputs are specified in appendix A. Marginals are log-uniform over the stated normalized ranges. The concave residual benefits are

$$E(x) = A_E(1 - e^{-x/e}), \quad R(u) = A_R(1 - e^{-u/r}), \quad I(v) = A_I(1 - e^{-v/i}). \tag{32}$$

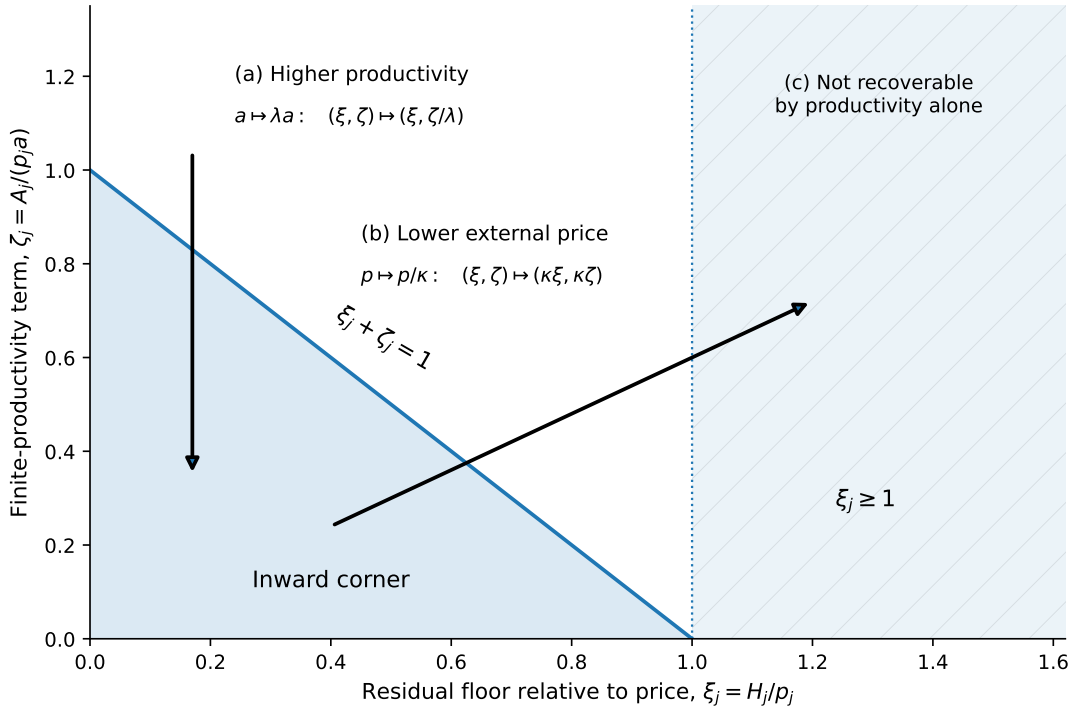


Figure 3: Exact logarithmic phase diagram and comparative statics for one external direction. The shaded triangle satisfies $H_j/p_j + A_j/(p_j a) \leq 1$; every external direction must satisfy this condition for the all-inward corner to be optimal. (a) Higher productivity reduces only ζ_j, producing a downward vertical move. (b) A price reduction to p_j/κ multiplies both coordinates by κ, producing a radial move away from the origin. (c) In the hatched region $\xi_j \geq 1$, no finite productivity restores the corner. For $A_j > 0$, $(1, 0)$ is only a limiting point. The arrows are schematic evaluations of the exact transformations; no archived inputs are overlaid and no population frequency is estimated.

The reference uses $(q, \epsilon_R, \epsilon_I, f) = (0.99, 0.01, 0.05, 1)$ and $m_a = m_r = 1$. Table 2 retains the full paired counts.

At instantaneous access, the reference has 219 inward corners; a uniform 10% price cut leaves 207. Only 364 reference inputs admit a finite productivity threshold with the same residual slopes; the other 660 violate a residual-floor condition. Perfect experiential substitution with zero residual weights yields 527 inward corners, while 497 still prefer external computational capacity. Raising only local productivity tenfold gives 314 inward corners; raising local and remote productivity together gives 222. At $f = 0.01$ and $f = 10^{-4}$, the count is zero. These are exact classifications of the archived inputs, not estimates of civilizational prevalence.

Eight additional scrambles of 4,096 points each give instantaneous inward fractions of 20.31–21.07% under the same log-uniform marginals. Linear-uniform marginals over the same endpoints give 10.99–11.82%. These 65,536 classifications use the exact boundary. Their variation describes sensitivity to the parameter-space measure, not observational uncertainty about the Galaxy.

5 Future resources can reverse a present inward preference

A one-horizon budget comparison omits one of the strongest reasons to expand: external industry can enlarge future resources. A minimal two-period construction makes this effect explicit while

also allowing local investment to be productive. Let $B_1 = 10$, $z_1 = B_1 - p_1 x$, and define

$$C_1(x) = C_{01} + a_1 \ln(1 + z_1) + r_1 x,$$
$$B_2(x) = B_{20} + g_L z_1 + g_X x, \tag{33}$$
$$W(x) = \ln[1 + C_1(x)] + \beta V_2[B_2(x)], \quad 0 \le x \le B_1/p_1. \tag{34}$$

Here $B_1 > 0$, $B_{20}, C_{01}, C_{02} \ge 0$, $a_1, a_2, p_1, p_2, r_1, r_2 > 0$, and $0 \le \beta \le 1$. The parameters $g_L, g_X \ge 0$ convert current allocations to next-period resources. Both directions receive a future-production channel. All represented downstream benefits of first-period projects are captured through B_2; this is not a full persistent-capital or interstellar replication model.

The continuation value is the second-period pure-computation problem,

$$V_2(B) = \max_{\substack{z_2, x_2 \ge 0 \\ z_2 + p_2 x_2 \le B}} \ln[1 + C_{02} + a_2 \ln(1 + z_2) + r_2 x_2]. \tag{35}$$

For $r_2 > 0$,

$$z_2^*(B) = \min\{B, \max(a_2 p_2 / r_2 - 1, 0)\}, \qquad x_2^*(B) = \frac{B - z_2^*(B)}{p_2}. \tag{36}$$

Its marginal value $V_2'(B)$ is the current best computational marginal return divided by $1 + C_2^*(B)$: $a_2/(1 + B)$ on the all-local branch, or r_2/p_2 on the mixed/external branch. The expressions agree at the switching budget. V_2 is increasing and concave.

Proposition 8 (A future-adjusted inward condition)**.** *For equations* (34) *and* (35)*, the all-inward first-period allocation is optimal if and only if*

$$\frac{r_1 - p_1 a_1/(1 + B_1)}{1 + C_{01} + a_1 \ln(1 + B_1)} + \beta V_2'(B_{20} + g_L B_1)(g_X - p_1 g_L) \le 0. \tag{37}$$

Proof. The first term in W is concave in x, and composition of concave V_2 with affine $B_2(x)$ is concave. For a differentiable concave function on an interval, the left endpoint is a global maximizer exactly when its right derivative is non-positive. Differentiating equation (34) at zero gives equation (37). □

The first term is the original within-period comparison. The second values the *difference* between external and local production of future resources. Counting g_X while setting g_L to zero without justification would bias the test against inward development; omitting g_X would bias it in favour. This extension is a structural check, not a calibrated decision rule for actual civilizations.

For $p_1 = p_2 = r_1 = r_2 = 1$, $a_1 = a_2 = 100$, $C_{01} = C_{02} = B_{20} = 1$, $g_L = 0.2$, and $\beta = 0.9$, present-only investment is all inward. The future-adjusted corner changes at $g_X = 0.4091$. At $g_X = 0.2$ both periods remain all inward. At $g_X = 2$, optimal first-period outward investment is $x = 3.7559$, leaving $z_1 = 6.2441$ locally and producing $B_2 = 9.7607$. All second-period investment is still local. Thus the optimal path can include *external investment undertaken to finance later inward activity.* This differs from claiming that external expansion is never useful. Figure 4 gives the full 201-setting construction.

The converse reversal is also possible. Keep $B_1 = 10$, $p_1 = p_2 = r_1 = r_2 = 1$ and $\beta = 0.9$, but set $a_1 = 5$, $a_2 = 100$, $C_{01} = C_{02} = 1$, $B_{20} = 0$, $g_L = 0.1$ and $g_X = 0$. The static term in equation (37) is $+0.03899$, and the present-only optimum is $x = 6$. The continuation term is -0.06310, making the total -0.02411. Concavity therefore makes $x = 0$ the two-period optimum. Local investment, rather than external investment, produces the more valuable continuation budget in this example. Supplementary Material section S3 reports scalar and joint-optimization checks of both cases.

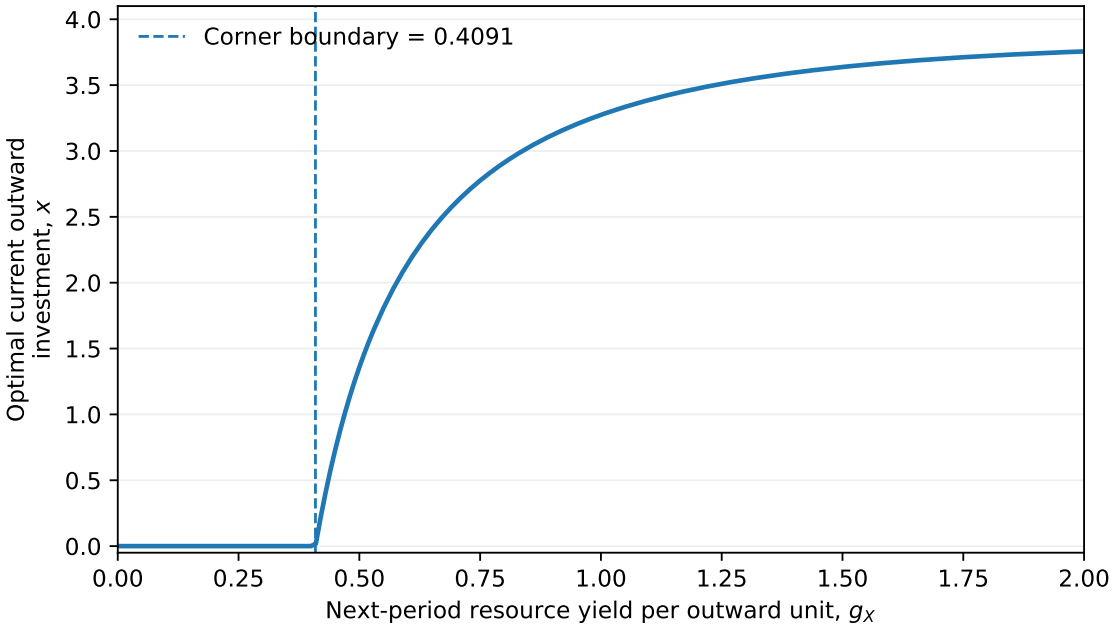


Figure 4: Two-period resource feedback. Local investment generates future resources at $g_L = 0.2$ per unit. Increasing the external yield g_X eventually overturns the all-inward present-only optimum. Parameters are normalized; the transition is an exact derivative boundary checked against a separate three-variable numerical optimization.

Thus static inwardness is neither sufficient nor necessary for dynamic inwardness: the $g_X = 2$ example disproves sufficiency, and the converse example disproves necessity. For these two examples, all second-period investment is local at the dynamic optimum. The same logical failures can be embedded in an infinite horizon by appending a summable discounted continuation whose budgets and returns are independent of the first two periods' allocations and whose subsequent optima are all inward. Such a continuation adds a constant to the current comparison and leaves its optimizer unchanged. This establishes the absence of a universal implication, not an infinite-horizon policy theorem for persistent capital or resource accumulation. Proposition 6 remains a characterization of repeated static choices; section 7.2 states the corresponding limitation.

6 From an inward civilization to a quiet galaxy

6.1 Initiation, survival, and arrival are not the same event

A civilization contains heterogeneous agents. A scalar objective in equation (12) is a representation of a specified decision process, not proof that every inhabitant shares its preferences. Recursive virtual population growth is especially relevant to this distinction. More virtual agents increase the potential expansionist population only when they possess physical agency, obtain access to actuators, or can create outward-capable descendants. Reliable containment might reduce that exposure, but it is not guaranteed by virtuality.

Let $a_i(t)$ be outward-capable autonomous nodes and $\lambda_i(t)$ their initiation rate per node-time. Under an independent Poisson approximation, the expected number of successful fronts from one civilization is

$$\mu = \sum_i \int \lambda_i(t) a_i(t) [1 - q_{R,i}(t)] \, \mathrm{d}t. \tag{38}$$

For a Galton–Watson process with offspring generating function G_R, the extinction probability is

the smallest solution $q_R = G_R(q_R)$; for Poisson offspring with mean R_{rogue}, $q_R = \exp[R_{\text{rogue}}(q_R - 1)]$ (Harris, 1963). Standard non-degenerate processes become extinct almost surely when $R_{\text{rogue}} \leq 1$. Survival for $R_{\text{rogue}} > 1$ is not equivalent to occupying the Galaxy or reaching Earth.

For a fixed independent list of civilizations with Poisson front means μ_i,

$$P_0^{\text{fixed}} = \exp\left(-\sum_i \mu_i\right). \tag{39}$$

If civilization origins instead form a Poisson process with intensity measure $\nu(\,\mathrm{d}z)$, independently marked by their front distributions, then

$$P_0^{\text{origins}} = \exp\left[-\int (1 - e^{-\mu(z)})\nu(\,\mathrm{d}z)\right]. \tag{40}$$

This follows from the Poisson generating functional (Last and Penrose, 2018). The fronts are clustered within their parent civilizations; replacing $1 - e^{-\mu}$ by μ is only a small-μ approximation. For $N = \bar{N} = 2$ and $\mu = 2$, the fixed-population and random-origin probabilities are approximately 0.0183 and 0.1774, respectively. Separate 250,000-realization simulations check these formulas.

To explain a null observation in a specified region by time T, replace $\mu(z)$ by the mean number $\mu_A(z, T)$ of front initiations that both survive and arrive under an explicit propagation model. Faster access can enlarge this arrival exposure even if typical investment remains inward. The allocation simulations do not calculate μ_A or a probability of galactic silence. Subcritical lineages can also produce finite local artefacts before extinction. Frequency among origins is therefore not frequency among occupied systems.

6.2 Visibility and non-identifiability

Settlement, industry, and observation indices do not determine power, spectrum, geometry, or duty cycle. A compact computational civilization may use substantial stellar power, while a wide probe network may be faint. The Dyson–Kardashev energy-processing tradition addresses observability but is not a direct scale of welfare or settlement demand (Dyson, 1960; Kardashev, 1964). Galaxy-scale infrared searches require a physical emission model and selection function (Wright et al., 2014a,b).

For mutually exclusive source classes with approximately independent Poisson detections,

$$\Lambda_j(T) = \int_0^T R_{\text{civ}}(t_e) S_j(T - t_e \mid t_e) \pi_j(T, t_e)\,\mathrm{d}t_e, \qquad P(0) = e^{-\sum_j \Lambda_j}. \tag{41}$$

Here S_j is a source-state probability and π_j a selection probability. Overlapping settlement, industry, and probe attributes require a union-detection probability rather than a sum that counts an object twice. Even in a stationary single-class reduction, a null result constrains the product $R_{\text{civ}}\tau\bar{\pi}$, not its factors separately. Multiplying the emergence rate by a factor and dividing detectability by the same factor leaves the likelihood unchanged when the parameters remain admissible. Non-detection alone cannot identify inward transition.

Recent observations illustrate this distinction. Project Hephaistos II identified seven follow-up candidates from approximately five million sources (Suazo et al., 2024). The cited 2026 JWST study attributes the excesses of two candidates to background galaxies (Zackrisson et al., 2026). A separate preprint tests warm technological components in 129 nearby galaxies without a preferred positive component (Curtis et al., 2026). These reports constrain the searched emission models; they do not establish virtual civilizations, universal low detectability, or the feasibility of local recursive growth. Solar-System artefact searches address another relevant class (Freitas, 1983; Haqq-Misra and Kopparapu, 2012).

The weaker settlement-composition branch suggests comparing resident infrastructure with automated industrial and information artefacts after correcting their selection functions. The wholly inward branch need not produce those distant artefacts. To discriminate the mechanism, a future study would need independently constrained resource budgets, support capacity, substitution behaviour, or a population of signatures for which intensive activity and outward acquisition can be estimated separately.

7 Verification, limitations, and interpretation

7.1 What is verified

Separate AI-assisted implementations reconstruct the equations without importing the generating model. The exact logarithmic test reproduces all 14,336 archived corner classifications; 11,002 tests check both sides of finite positive productivity thresholds. The stored optimized shares also pass objective, feasibility and concavity-gap checks. Scalar and joint optimizations check the curvature examples, production alternative, pure-computation budget solution and two-period extension; direct accounting checks the workload examples, including the memory failure beyond the stated horizon. Complete proofs establish the mathematical claims; these numerical tests check implementations and examples, not extraterrestrial premises. Supplementary Material sections S1–S3 provide certificate details, executable commands, numerical outputs and the seed-versus-`rng` comparison. Frozen inputs, environment versions and integrity manifests accompany the source.

7.2 Limitations that change the conclusion

Physical support and recursive interpretation. The workload frontier is a homogeneous service model, not an engineered architecture or a test of consciousness. Efficiency or fidelity may plateau, stored state may dominate memory, and digital population growth may restore scarcity. Per-agent resources increase only if useful computational growth outpaces agent growth. The fixed-budget joint witness chooses rather than derives its productivity path. No engineering model here establishes indefinite growth of every effective support capacity. Recursion is neither necessary nor sufficient for inward allocation.

Preferences, production and mathematical scope. The objective represents a specified decision process, not a shared utility function of every inhabitant. Divisible investment, positive prices, differentiability and concavity are substantive restrictions. The local logarithm and external production functions are not empirically identified; lemma 7 separates corner robustness from interior-allocation sensitivity. The value-curvature parameter also changes the conclusion, as proposition 5 shows. Small residual utility weights do not certify physical safety or remove the value of geographically independent backups. Fixed project sizes, start-up thresholds and increasing returns can invalidate a local test. For example,

$$U(x) = \ln\{1 + 2\ln(2 - x) + 0.1x\} + 3x^2, \qquad 0 \le x \le 1, \tag{42}$$

with $z = 1 - x$, has $U'(0) \simeq -0.3772$, yet $U(1) \simeq 3.0953 > U(0) \simeq 0.8697$. This violates the concavity assumption rather than proposition 2. Under added hard constraints, necessity of the corner inequalities additionally requires the proposed small budget-transfer directions to remain feasible.

Persistence and future opportunities. Proposition 6 characterizes a sequence of static optima, not an infinite-horizon policy. Its conditions are neither necessary nor sufficient, in general, for an all-inward optimal policy in a dynamic model: section 5 gives both reversals

in two periods and explains their extension when the later continuation is independent. A substantive infinite-horizon criterion would require specified state dynamics, feasible policies, and a well-defined discounted or long-run objective. With logarithmic value, fixed positive residual slopes and bounded prices exclude an all-inward path when budgets become unbounded. Residual decay or sufficiently rising relative prices must satisfy the exact productivity condition as well; simply observing that $S_j(t) \to 0$ is insufficient. The two-period model includes both local and external future resource production, but not a full ecology of accumulated capital, strategic competition, irreversible settlement, survival or real options.

Access and physical assumptions. The access-cost map is a sensitivity family, not a propulsion model. Proposed warp geometries do not establish operational faster-than-light travel (Alcubierre, 1994); energy-condition and causality objections remain relevant (Everett, 1996; Santiago et al., 2022). The counterfactual limiting cases assume consistent scheduling without usable travel into one's own past. A rapid material courier may also carry information. Faster access can increase both external opportunities and arrival exposure, not merely reduce travel time.

Inference about populations. The synthetic design is not an astronomical prior. Neither its counts nor variation across scrambles estimates the prevalence of inward civilizations. Allocation does not identify power use, spectra, duty cycles, autonomous expansionist exposure or arrival probabilities. Survey nulls are therefore not selective confirmation of CITH. Empirical application would require independent evidence for relevant resource and preference parameters, a propagation model and comparative observational likelihoods.

8 Conclusion

The relevant possibility is a growing civilization, not necessarily a satiated or inactive one. Internal computational uses, including recursively generated virtual worlds, can compete successfully with additional external projects. The resource frontier and the common-budget model make the two requirements distinct: the chosen workload must remain physically supportable, and inward investment must remain preferable at the relevant opportunity costs.

Continued internal growth with no additional external investment is therefore conditional on a path, not established by a favourable corner at one budget. Under logarithmic computational value, budgets tending to infinity, fixed positive residual external slopes and bounded relative prices are incompatible with persistent all-inward allocation, irrespective of productivity. Persistence requires the price-to-residual-slope ratio to exceed $(1 + B)\ln(1 + B)$ wherever a residual slope is positive, together with the exact productivity threshold and continuing local feasibility. The curvature extension further shows that the finite residual floor is specific to logarithmic value: slower marginal-value decay permits an eventual productivity threshold at fixed budgets and prices, whereas faster decay makes sufficiently high productivity incompatible with inwardness when residual slopes are positive.

Concave external alternatives preserve the corner comparison when initial returns are unchanged, but alter mixed allocations. Cheap external access and future resource production supply additional reasons to expand. These boundaries define a quantitatively specified alternative to inferring inevitable expansion from capability alone. They do not establish its prevalence, indefinite engineering realization, or a population-level explanation of galactic silence.

Declarations

Data and code availability. All model data are normalized synthetic inputs. The accompanying source-and-replication package contains the frozen allocation inputs and optimized solutions,

separate equation-level audit code, workload and budget checks, full proofs, new curvature and pathwise tests, figure arrays, environment records, and SHA-256 manifests. Reproduction commands and the exact Sobol generator call are given in appendix A and the Supplementary Material. No public repository DOI is asserted; these materials should accompany submission.

Author information. Hyeonjun Joo is an independent researcher in the Republic of Korea. Correspondence: jonejoo0805@gmail.com.

Financial support and competing interests. The author reports no specific funding and no competing interests.

Author contributions. Hyeonjun Joo: Conceptualization; Methodology; Formal analysis; Investigation; Visualization; Writing; Project administration. The author is responsible for approving the scientific interpretation and final manuscript.

Use of artificial-intelligence tools. OpenAI ChatGPT assisted mathematical model development, code generation, computational checks, literature checking, language revision, and LaTeX preparation, including the work dated 12 October 2026. Both the generating and checking implementations were AI-assisted. Separate implementations do not constitute independent human peer review. All plotted values come from archived calculations. The human author retains responsibility for the scientific claims, references, numerical results, final wording, and submission decision. No AI system is an author.

Ethics. No human participants, animals, personal research data, or identifiable-person observational data were studied.

A Numerical specification and paired results

The canonical unit points and transformed parameters are the arrays `sobol_vectors` and `parameters` in `data/allocation/design.npz`. The generating environment records Python 3.13.5, NumPy 2.3.5 and SciPy 1.17.0. The exact call is

```
from scipy.stats import qmc
u = qmc.Sobol(d=14, scramble=True,
              seed=20260905).random_base2(m=10)
```

The default `bits=None` resolves to 30 bits; optimization is `None`. No points are skipped or thinned. For the ordered endpoint vectors in table 1, the transformation is

```
lo = np.log(lower_endpoints)
hi = np.log(upper_endpoints)
parameters = np.exp(lo + u * (hi - lo))
```

Every scenario uses the same transformed rows.

Re-execution in SciPy 1.17.0 reproduces the archived unit points bit for bit. Floating-point exponentiation changes some transformed entries by at most a relative 2.22×10^{-16} in the present environment; all classifications agree. Evaluation of the frozen arrays reproduces 219 instantaneous-reference inward corners and 527 under perfect substitution with zero residual weights, as well as all other counts in table 2.

With `seed=20260905`, the archived unit points are reproduced bit for bit under SciPy 1.17.0 and 1.18.1; `rng=20260905` yields a different scramble, with 208 instantaneous and 530 perfect-substitution inward corners. The frozen arrays, not the seed alone, define the reference experiment. Supplementary Material section S2 records the generator specification and file checksums.

The parameter order is $(B, C_0, a, r_n, r_h, A_E, A_R, A_I, e, r, i, p_{n0}, p_{h0}, p_{m0})$. All base marginal ranges are log-uniform in the reference experiment. Values of e, r, i in this appendix are return scales, not energy, radius, or individual identifiers.

Table 1: Normalized ranges in the 14-dimensional reference design.

Quantity	Range	Meaning
B	1–100	common investment budget
C_0	0.1–10	existing useful computational service
a	1–1000	intensive computational coefficient
r_n, r_h	0.1–10	external computational returns
A_E, A_R, A_I	0.1–10	benefit amplitudes
e, r	1–30	experiential and resilience scales
i	1–100	information scale
p_{n0}	1–10	resident unit price
p_{h0}	0.1–3	industry-only unit price
p_{m0}	0.01–0.3	observation-only unit price

The sublight scenario has $(s_t, s_c) = (0.01, 1)$, the communication-only variant $(0.01, \infty)$, the light-speed case $(1, 1)$, and combined faster-access cases $(10, 10)$, $(1000, 1000)$, and (∞, ∞). The no-substitution variant sets $q = 0$. Large residual values set $\epsilon_R = \epsilon_I = 1$, either retaining $q = 0.99$ or additionally setting $q = 0$. The two tenfold productivity variants set $(m_a, m_r) = (10, 1)$ and $(10, 10)$. The perfect-substitution variant sets $(q, \epsilon_R, \epsilon_I) = (1, 0, 0)$. Reduced-price variants set $f = 0.01$ or 10^{-4} with instantaneous access; other parameters remain at reference settings. A material-speed variable is not allowed to make numerical prices exactly zero.

Table 2: All-inward inputs out of 1,024, with a conditional price-robustness check. The last column applies a uniform 10% price reduction to each listed scenario while retaining its other settings. It is computed from the exact boundary, not from a new behavioral assumption or an observational sample.

Scenario	All inward	Still inward after 10% price cut
Sublight material travel	929	923
Fast communication only	929	923
Light-speed limit	326	313
Travel and communication at 10c	234	219
Travel and communication at 1000c	219	207
Instantaneous reference	219	207
No experiential substitution	146	135
Large residual values	14	14
No substitution + large residuals	11	10
Local productivity ×10	314	296
Local and remote productivity ×10	222	211
Perfect substitution; no residual values	527	505
Outward price floor ×0.01	0	0
Outward price floor ×0.0001	0	0

Corner classification uses the exact inequalities, not optimizer-selected shares. The stored optimal shares are checked separately using the objective and concavity certificates described in Supplementary Material section S1. Small platform-dependent changes in transformed

parameters or nearly tied optimized shares do not change any archived corner classification in the checks reported here.

For design-measure sensitivity, Sobol seeds 20260911 through 20260918 each generate 2^{12} points. Both log-uniform and linear-uniform transformations use the same bounds in table 1. Repeated scrambles assess numerical integration stability under a selected measure. They do not supply a prior justified by extraterrestrial observations.

References

Miguel Alcubierre. The warp drive: hyper-fast travel within general relativity. *Classical and Quantum Gravity*, 11:L73–L77, 1994. doi: 10.1088/0264-9381/11/5/001.

Stuart Armstrong and Anders Sandberg. Eternity in six hours: Intergalactic spreading of intelligent life and sharpening the Fermi paradox. *Acta Astronautica*, 89:1–13, 2013. doi: 10.1016/j.actaastro.2013.04.002.

Charles H. Bennett. Logical Reversibility of Computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973. doi: 10.1147/rd.176.0525.

Charles H. Bennett, Robin Hanson, and C. Jess Riedel. Comment on 'The Aestivation Hypothesis for Resolving Fermi's Paradox'. *Foundations of Physics*, 49:820–829, 2019. doi: 10.1007/s10701-019-00289-5. arXiv:1902.06730.

Nick Bostrom. Astronomical Waste: The Opportunity Cost of Delayed Technological Development. *Utilitas*, 15(3):308–314, 2003. doi: 10.1017/S0953820800004076.

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. doi: 10.1017/CBO9780511804441. URL `https://web.stanford.edu/~boyd/cvxbook/`.

G. D. Brin. The "Great Silence": the controversy concerning extraterrestrial intelligent life. *Quarterly Journal of the Royal Astronomical Society*, 24(3):283–309, 1983.

Jonathan Carroll-Nellenback, Adam Frank, Jason T. Wright, and Caleb Scharf. The Fermi Paradox and the Aurora Effect: Exo-civilization Settlement, Expansion, and Steady States. *The Astronomical Journal*, 158(3):117, 2019. doi: 10.3847/1538-3881/ab31a3.

Milan M. Ćirković and Robert J. Bradbury. Galactic Gradients, Postbiological Evolution and the Apparent Failure of SETI. *New Astronomy*, 11(8):628–639, 2006. doi: 10.1016/j.newast.2006.04.003.

Olivia Curtis, Aidan J. Rowland, Jason T. Wright, Caryl Gronwall, Jakob M. Helton, and Joel Leja. The $\hat{\mathrm{G}}$ Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies. V. When Galaxies Glow with Industry, 2026. Preprint; arXiv:2608.12458.

Steven J. Dick. Cultural Evolution, the Postbiological Universe and SETI. *International Journal of Astrobiology*, 2(1):65–74, 2003. doi: 10.1017/S147355040300137X.

Freeman J. Dyson. Search for Artificial Stellar Sources of Infrared Radiation. *Science*, 131(3414):1667–1668, 1960. doi: 10.1126/science.131.3414.1667.

Allen E. Everett. Warp drive and causality. *Physical Review D*, 53:7365–7368, 1996. doi: 10.1103/PhysRevD.53.7365.

Robert A. Freitas. The Search for Extraterrestrial Artifacts (SETA). *Journal of the British Interplanetary Society*, 36:501–506, 1983.

Robin Hanson, Daniel Martin, Calvin McCarter, and Jonathan Paulson. If Loud Aliens Explain Human Earliness, Quiet Aliens Are Also Rare. *The Astrophysical Journal*, 922(2):182, 2021. doi: 10.3847/1538-4357/ac2369.

Jacob Haqq-Misra and Ravi Kumar Kopparapu. On the Likelihood of Non-terrestrial Artifacts in the Solar System. *Acta Astronautica*, 72:15–20, 2012. doi: 10.1016/j.actaastro.2011.10.010.

Jacob D. Haqq-Misra and Seth D. Baum. The Sustainability Solution to the Fermi Paradox. *Journal of the British Interplanetary Society*, 62:47–51, 2009. doi: 10.48550/arXiv.0906.0568. arXiv:0906.0568.

Theodore E. Harris. *The Theory of Branching Processes*. Springer, Berlin, 1963.

Michael H. Hart. An Explanation for the Absence of Extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16:128–135, 1975.

Sergey Ivliev. Autonomous AI-Cosmoindustry and the Quiet Expansion Filter: A Threshold-Based Resolution of the Fermi Paradox, 2026. Preprint; arXiv:2606.13914.

Chris J. Jackson and Christian Criado-Perez. Why the Fermi paradox may not be well explained by Wong and Bartlett's theory of civilization collapse. A Comment on: 'Asymptotic burnout and homeostatic awakening: a possible solution to the Fermi paradox?' (2022) by Wong and Bartlett. *Journal of the Royal Society Interface*, 21(219):20240140, 2024. doi: 10.1098/rsif.2024.0140.

Nikolai S. Kardashev. Transmission of Information by Extraterrestrial Civilizations. *Soviet Astronomy*, 8:217–221, 1964.

David Kipping. The Cosmological Hart–Tipler Conjecture, 2026. Preprint; arXiv:2606.04044.

Rolf Landauer. Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

Günter Last and Mathew Penrose. *Lectures on the Poisson Process*. Cambridge University Press, 2018. doi: 10.1017/9781316104477.

Seth Lloyd. Ultimate Physical Limits to Computation. *Nature*, 406:1047–1054, 2000. doi: 10.1038/35023282.

William I. Newman and Carl Sagan. Galactic Civilizations: Population Dynamics and Interstellar Diffusion. *Icarus*, 46(3):293–327, 1981. doi: 10.1016/0019-1035(81)90135-4.

S. Jay Olson. Homogeneous Cosmology with Aggressively Expanding Civilizations. *Classical and Quantum Gravity*, 32(21):215025, 2015. doi: 10.1088/0264-9381/32/21/215025.

Rodolfo Pitti. The Inward Transition and the Detectability Window: Thermodynamic Constraints on Extraterrestrial Signal Detection in Post-AGI Civilizations. Non-peer-reviewed working paper, January 2026. URL `https://www.academia.edu/161276613/The_Inward_Transition_and_the_Detectability_Window_Thermodynamic_Constraints_on_Extraterrestrial_Signal_Detection_in_Post_AGI_Civilizations`.

Anders Sandberg, Stuart Armstrong, and Milan M. Ćirković. That Is Not Dead Which Can Eternal Lie: The Aestivation Hypothesis for Resolving Fermi's Paradox, 2017. arXiv:1705.03394.

Jessica Santiago, Sebastian Schuster, and Matt Visser. Generic warp drives violate the null energy condition. *Physical Review D*, 105:064038, 2022. doi: 10.1103/PhysRevD.105.064038.

Alexander Vladimirovich Shkurko. The Social Science Perspective on the Fermi Paradox. *International Journal of Astrobiology*, 23:e13, 2024. doi: 10.1017/S1473550424000089.

John M. Smart. The Transcension Hypothesis: Sufficiently Advanced Civilizations Invariably Leave Our Universe, and Implications for METI and SETI. *Acta Astronautica*, 78:55–68, 2012. doi: 10.1016/j.actaastro.2011.11.006.

Matías Suazo, Erik Zackrisson, Priyatam K. Mahto, Fabian Lundell, Carl Nettelblad, Andreas J. Korn, Jason T. Wright, and Suman Majumdar. Project Hephaistos – II. Dyson Sphere Candidates from Gaia DR3, 2MASS, and WISE. *Monthly Notices of the Royal Astronomical Society*, 531(1):695–707, 2024. doi: 10.1093/mnras/stae1186. arXiv:2405.02927.

Frank J. Tipler. Extraterrestrial Intelligent Beings Do Not Exist. *Quarterly Journal of the Royal Astronomical Society*, 21:267–281, 1980.

Michael L. Wong and Stuart Bartlett. Asymptotic Burnout and Homeostatic Awakening: A Possible Solution to the Fermi Paradox? *Journal of the Royal Society Interface*, 19(190): 20220029, 2022. doi: 10.1098/rsif.2022.0029.

Jason T. Wright, Brendan Mullan, Steinn Sigurdsson, and Matthew S. Povich. The $\hat{\mathrm{G}}$ Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies. I. Background and Justification. *The Astrophysical Journal*, 792(1):26, 2014a. doi: 10.1088/0004-637X/792/1/26.

Jason T. Wright, Roger L. Griffith, Steinn Sigurdsson, Matthew S. Povich, and Brendan Mullan. The $\hat{\mathrm{G}}$ Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies. II. Framework, Strategy, and First Result. *The Astrophysical Journal*, 792(1):27, 2014b. doi: 10.1088/0004-637X/792/1/27.

Erik Zackrisson, Arjan Bik, Anita Ali Asgar, Olivia Curtis, Jason T. Wright, Tongtian Ren, Roberto J. Assef, Andrew Blain, Alexis Brandeker, Michael A. Garrett, Uma Gorti, Andreas J. Korn, Priyatam K. Mahto, and Armin Nabizadeh. Project Hephaistos – IV. James Webb Space Telescope Observations of Two Dyson Sphere Candidates, 2026. Preprint; arXiv:2607.09460.